Chapter 1

Limits and Continuity

In this chapter, we move on from thinking about *sequences*, which are functions $\mathbb{N} \to \mathbb{R}$, and we start to build some theory for functions $\mathbb{R} \to \mathbb{R}$, in particular what constitutes a "*nice*" function.

What motivates this chapter is that a "*nice*" function should be able to drawn without taking our pen off the paper, because we expect nice properties of such functions; this is intuition is a property called the "*Intermediate Value Property*", and does indeed hold for *continuous* functions (but is not exclusive to continuous functions).

However, the above discussion is by no means a definition. We will define what we mean by *continuity* of a function $f : \mathbb{R} \to \mathbb{R}$ once we understand what a *limit* of a function is. In particular, since we are working with functions whose domain is $\mathbb{R}$ now, the notion of *limit* can be made meaningful without sending $x \to \infty$ (i.e. we can define the limit of a function $f(x)$, say, as $x \to a$ where $a \in \mathbb{R}$ is some real number).

1.1 Limits of Functions

We wish to adapt the definition of limit for functions $\mathbb{R} \to \mathbb{R}$. First, we make the definition for limits as $|x| \to \infty$, which are analogous to the definition of a limit for sequences (but of course, rather than only being able to send n to positive infinity like in sequences, there is another direction, the negative numbers, so we can also define limits as $x \to -\infty$ as well).

Definition 1.1. Let $f : \mathbb{R} \to \mathbb{R}$ be a function, and let $L \in \mathbb{R}$.

1. We say that $f(x)$ tends to L as $x \to \infty$ if

$$\forall \varepsilon > 0, \exists R \in \mathbb{R}, \forall x \geq R, |f(x) - L| < \varepsilon.$$

2. We say that $f(x)$ tends to L as $x \to -\infty$ if

$$\forall \varepsilon > 0, \exists R \in \mathbb{R}, \forall x \leq R, |f(x) - L| < \varepsilon.$$

Definition 1.2. Let $f : \mathbb{R} \to \mathbb{R}$ be a function.

1. We say that $f(x)$ tends to ∞ as $x \to \infty$ if
$$\forall M > 0, \exists R \in \mathbb{R}, \forall x \geq R, f(x) > M.$$
2. We say that $f(x)$ tends to ∞ as $x \to -\infty$ if
$$\forall M > 0, \exists R \in \mathbb{R}, \forall x \leq R, f(x) > M.$$

There is an obvious variation of the above definitions for the definitions of $\lim_{x\to\infty} f(x) = -\infty$ and $\lim_{x\to-\infty} f(x) = -\infty$.

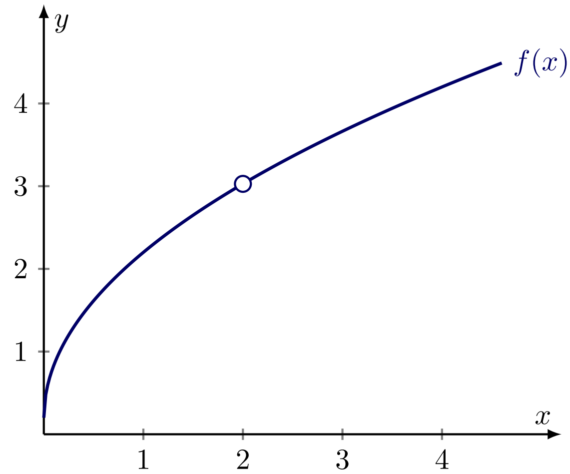
To define "$\lim_{x\to a} f(x) = L$", we need to be precise about what "for x close enough to a" actually means. Also, it is going to matter whether we approach a from the side $x > a$ or from the side $x < a$, so writing $\lim_{x\to a} f(x) = L$ may not always make sense.

Definition 1.3. Let $a \in \mathbb{R}$ and let $f : (a, \infty) \to \mathbb{R}$ be a function. We say that $f(x)$ tends to $L \in \mathbb{R}$ as $x \to a^+$ if
$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in (a, \infty), x \in (a, a + \delta) \implies |f(x) - L| < \varepsilon.$$

Above, we formalised "for $x > a$ sufficiently close to a" as "$x \in (a, a + \delta)$ for some $\delta > 0$".

Definition 1.4. Let $a \in \mathbb{R}$ and let $f : (b, a) \to \mathbb{R}$ be a function for some $b < a$. We say that $f(x)$ tends to $L \in \mathbb{R}$ as $x \to a^-$ if
$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in (b, a), x \in (a - \delta, a) \implies |f(x) - L| < \varepsilon.$$

Remark. Limits are entirely about the behaviour of f *near* a, not *at* a. The value $f(a)$ does not even need to be defined. If $\lim_{x\to a} f(x) = L$ exists but $f(a)$ is undefined or doesn't equal L, we can define a "continuous extension" $\tilde{f}$ where $\tilde{f}(x) = f(x)$ for $x \neq a$ and $\tilde{f}(a) = L$. This is a useful trick for when we want to apply some stronger theorems which only apply to, say, continuous functions.



For example, in the picture above, the function $f(x)$ has $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x) = 3$, but is not defined at $x = 2$. If we defined $f(2) = 3$, then f would be a continuous function.

Lemma 1.5. *Suppose $f : (a, b) \to \mathbb{R}$ is a function, and we have*

$$\lim_{x\to a^+} f(x) = L_1 \quad \textit{and} \quad \lim_{x\to a^+} f(x) = L_2.$$

Then $L_1 = L_2$.

Proof. Let $\varepsilon > 0$ be arbitrary. Since $\lim_{x\to a^+} f(x) = L_1$, we can let $\delta_1 > 0$ be such that for any $x \in (a, a + \delta_1)$, $|f(x) - L_1| < \frac{\varepsilon}{2}$. Similarly, since $\lim_{x\to a^+} f(x) = L_2$, we can let $\delta_2 > 0$ be such that for any $x \in (a, a + \delta_2)$, $|f(x) - L_2| < \frac{\varepsilon}{2}$. Let $\delta = \min\{\delta_1, \delta_2\}$ and pick $x \in (a, a + \delta)$. By the triangle inequality we have

$$|L_1 - L_2| = |L_1 - f(x) + f(x) - L_2| \leq |f(x) - L_1| + |f(x) - L_2| = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

so since $\varepsilon > 0$ was arbitrary, it follows that $L_1 = L_2$. □

This lemma holds if we replace $x \to a^+$ with $x \to a^-$ and the proof is identical. Note that the above fact does not say that we must have $\lim_{x\to a^-} f(x) = \lim_{x\to a^+}$, even if both exist.

Now we will define a two-sided limit; note that this requires both the left and right limits to exist and $\lim_{x\to a^-} f(x) = \lim_{x\to a^+}$. The function must also be defined on an interval of the form $(b, c) \setminus \{a\}$ for some $b < a < c$ (notice again that $f(a)$ need not be defined for this definition to make sense):

Definition 1.6. Let $a \in \mathbb{R}$ and let $f : (b, c) \setminus \{a\} :\to \mathbb{R}$ be a function for some $b < a < c$. We say that $f(x)$ tends to $L \in \mathbb{R}$ as $x \to a$ if

$$\lim_{x\to a^-} f(x) = L \quad \text{and} \quad \lim_{x\to a^+} f(x) = L.$$

In other words, $\lim_{x\to a} f(x) = L$ if

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x \in (b, c) \setminus \{a\},\ 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

Definition 1.7. Let $I \subset \mathbb{R}$. We say that I is an *open interval* if it is a set of the form:

- an interval (a, b) for some $a < b$ in $\mathbb{R}$,
- an interval $(-\infty, b)$,
- an interval (a, ∞),
- $I = \mathbb{R}$.

Theorem 1.8 (Algebra of Limits)**.** *Let $I \subset \mathbb{R}$ be an open interval, and let $a \in I$. If* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} g(x) = M$*, then:*

1. $\lim_{x\to a}(f(x) + g(x)) = L + M$*;*
2. $\lim_{x\to a}(f(x)g(x)) = LM$*;*
3. $\lim_{x\to a} \frac{1}{f(x)} = \frac{1}{L}$ *(provided $L \neq 0$ and $f(x) \neq 0$ near a, except possibly at a).*

Proof. Exercise (compare with the proofs for the Algebra of Limits for sequences). □

Remark. Let $I \subset \mathbb{R}$ be an interval, and let $a \in I$. When we say a property holds "near a", we mean there exists some $\delta > 0$ such that the property holds for all $x \in (a-\delta, a+\delta) \cap I$.

When we append the phrase "except possibly at a", we restrict x to the punctured neighborhood $0 < |x-a| < \delta$. Therefore, when talking about limits, we can concisely discuss properties of functions that only need to hold *locally.* For example, a common requirement for theorems which involve limits of functions like $\frac{1}{f(x)}$ is that $f(x) \neq 0$ near a, except possibly at a.

1.2 Continuity

Definition 1.9 (Continuity at a point)**.** Let $I \subset \mathbb{R}$ be an open interval, and let $f : I \to \mathbb{R}$ be a function. We say that f is *continuous* at $a \in I$ if

$$\lim_{x\to a} f(x) = f(a).$$

Equivalently, f is continuous at $a \in I$ if

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in I, |x-a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

Remark. Note that we do not need to specify that $|x-a| > 0$ since we are assuming f is defined at a now, and if $|x-a| = 0$ then $x = a$ and the inequality $|f(a) - f(a)| < \varepsilon$ is trivially satisfied for any $\varepsilon > 0$.

Exercise. Negate the above string of quantifiers to get a definition for *discontinuity* at a point $a \in I$.

Definition 1.10. Let $I \subset \mathbb{R}$ be an open interval. We say that $f : I \to \mathbb{R}$ is *continuous everywhere*, or *continuous*, if f is continuous at a for all $a \in I$.

Example 1.11.

1. Constant functions are continuous everywhere. Indeed, fix $\lambda \in \mathbb{R}$, and define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = \lambda$ for all $x \in \mathbb{R}$. Let $\varepsilon > 0$ and $a \in \mathbb{R}$ be arbitrary. Choose $\delta = 1$, and let $x \in \mathbb{R}$ satisfy $|x-a| < \delta$. Then we have

$$|f(x) - f(a)| = |\lambda - \lambda| = 0 < \varepsilon,$$

so f is continuous everywhere.

2. The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x$ is continuous everywhere. Indeed, let $\varepsilon > 0$ and $a \in \mathbb{R}$ be arbitrary. Choose $\delta = \varepsilon$ and let $x \in \mathbb{R}$ be such that $|x - a| < \delta$. Then we have
$$|f(x) - f(a)| = |x - a| < \delta = \varepsilon,$$
so f is continuous everywhere.

3. The function $f : \mathbb{R} \to \mathbb{R}$ defined by
$$f(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$
is continuous everywhere except at $a = 0$. First we show discontinuity at $a = 0$. Pick $\varepsilon = \frac{1}{2}$, and let $\delta > 0$ be arbitrary. We know that $f(a) = 1$, so we let $x = -\frac{\delta}{2} \in (-\delta, \delta)$. Then we have
$$|f(x) - f(0)| = |0 - 1| = 1 > \frac{1}{2} = \varepsilon,$$
so f is discontinuous at $a = 0$

Now, let $\varepsilon > 0$ and $c \neq 0$ be arbitrary. Choose $\delta = \frac{|c|}{2}$, and let $x \in \mathbb{R}$ be such that $|x - c| < \delta$. Then in particular,
$$\frac{|c|}{2} < |x| < \frac{3|c|}{2},$$
so if $x < 0$, then $f(x) = f(c) = 0$, and if $x > 0$, then $f(x) = f(c) = 1$, so in either case $|f(x) - f(c)| = 0 < \varepsilon$, so f is continuous at c for every $c \neq 0$.

1.2.1 Properties of Continuous Functions

Theorem 1.12 (Algebra for Continuous Functions)**.** *Let $I \subset \mathbb{R}$ be an open interval and let $f, g : I \to \mathbb{R}$ be continuous. Then*

1. *$f + g : I \to \mathbb{R}$ is continuous;*
2. *$fg : I \to \mathbb{R}$ is continuous;*
3. *If $f(x) \neq 0$ for all $x \in I$ then the function $\frac{1}{f} : I \to \mathbb{R}$ is continuous.*

Proof. All items follow from the Algebra of Limits for functions (Theorem 1.8). □

Example 1.13. In Example 1.11, we showed that the identity function on $\mathbb{R}$ is continuous, and constant functions on $\mathbb{R}$ are continuous. Therefore, by induction, for every $n \in \mathbb{N}$, $x \mapsto a_n x^n$ where $a_n \in \mathbb{R}$ is a constant is continuous on $\mathbb{R}$ using item 2 in the Algebra for Continuous functions (Theorem 1.12). Using item 1 in the Algebra for Continuous Functions inductively, we deduce that *polynomials* are continuous, i.e. functions $P : \mathbb{R} \to \mathbb{R}$ of the form
$$P(x) = a_0 + a_1 x + \cdots + a_{n-1} x^{n-1} + a_n x^n,$$
where $a_0, \ldots, a_n \in \mathbb{R}$ with $a_n \neq 0$ are constants, are continuous.

Furthermore, using this fact, by combining items 2 and 3 in the Algebra for Continuous functions we also deduce that functions of the form

$$R : I \to \mathbb{R}, \quad x \mapsto \frac{P(x)}{Q(x)},$$

where P, Q are polynomials and $I \subset \mathbb{R}$ is an open interval on which Q never vanishes, are continuous. Such a function is called a *rational function.*

Theorem 1.14. *Let $I \subset \mathbb{R}$ be an open interval, and let $f : I \to \mathbb{R}$ be continuous. Let $J \subset \mathbb{R}$ be an open interval such that $J \supset f(I)$, and let $g : J \to \mathbb{R}$ be continuous. Then*

$$g \circ f : I \to \mathbb{R}$$

is continuous.

Proof. Let $\varepsilon > 0$ and $a \in I$ be arbitrary. Since $g : J \to \mathbb{R}$ is continuous at $f(a) \in J$, there exists $\delta > 0$ such that for all $y \in J$, if $|y - f(a)| < \delta$ then $|g(y) - g(f(a))| < \varepsilon$. Since $f : I \to \mathbb{R}$ is continuous at $a \in I$, there exists $\delta' > 0$ such that for all $x \in I$, if $|x - a| < \delta'$ then $|f(x) - f(a)| < \delta$. Therefore, we have, for $x \in I$ with $|x - a| < \delta'$, $|f(x) - f(a)| < \delta$, so it follows that (with $y = f(x)$),

$$|g(f(x)) - g(f(a))| < \varepsilon,$$

so $g \circ f$ is continuous. □

Example 1.15. Using the above Theorem 1.14 and Example 1.13 (where we deduced the continuity of polynomials and rational functions on $\mathbb{R}$), we can deduce the continuity of a large class of functions without having to use the ε-δ definition. For example, if for the moment we assume the continuity of the functions sin, cos and exp (which we will show in later chapters), we can deduce that

$$x \mapsto \sin\left(\frac{x}{x^2+1}\right), \quad x \mapsto \sum_{k=1}^{n} \frac{1}{k\pi} \cos(kx), \quad x \mapsto \exp\left(-x^2\right)$$

are all continuous on $\mathbb{R}$.

1.2.2 Sequential Continuity

We can translate problems about continuity back into the familiar world of sequences with the following concept (which is a bit more general than the ε-δ definition):

Definition 1.16 (Sequential Continuity)**.** Let $I \subset \mathbb{R}$ be an open interval. A function $f : I \to \mathbb{R}$ is *sequentially continuous* at $a \in I$ if for any sequence (x_n) contained in I with $\lim_{n\to\infty} x_n = a$, we have $\lim_{n\to\infty} f(x_n) = f(a)$.

Remark. We say that a function $f : I \to \mathbb{R}$, where $I \subset \mathbb{R}$ is an open interval, is *sequentially continuous* if it is sequentially continuous at every $a \in I$.

In $\mathbb{R}$ (and, in fact, in any metric space) the concepts of sequential continuity and ε-δ continuity are logically equivalent:

Theorem 1.17 (Sequential Characterisation of Continuity). *Let $I \subset \mathbb{R}$ be an open interval and let $f : I \to \mathbb{R}$ be a function. Then f is continuous at $a \in I$ if and only if f is sequentially continuous at $a \in I$.*

Proof. Suppose f is continuous at $a \in I$, and let $\varepsilon > 0$ be arbitrary. By definition, we can choose $\delta > 0$ such that for all $x \in I$ such that $|x - a| < \delta$, $|f(x) - f(a)| < \varepsilon$. Since $x_n \to a$ as $n \to \infty$, we can let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|x_n - a| < \delta$. Therefore, $|f(x_n) - f(a)| < \varepsilon$ for $n \geq N$, so $f(x_n) \to f(a)$ as $n \to \infty$.

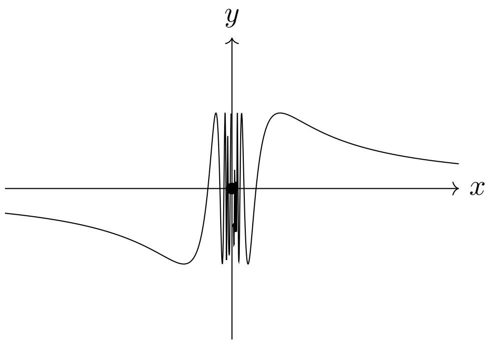
Conversely, suppose f is sequentially continuous at $a \in I$, and suppose for contradiction that f is not continuous at $a \in I$. Therefore, there exists some $\varepsilon_0 > 0$ such that for all $\delta > 0$, there exists $x \in I$ such that $|x - a| < \delta$ and $|f(x) - f(a)| \geq \varepsilon_0$. In particular, for $n \in \mathbb{N}$ we can let $\delta_n := \frac{1}{n}$, and by definition we can pick $x_n \in I$ with $|x_n - a| < \delta_n$ such that $|f(x_n) - f(a)| \geq \varepsilon_0$; by induction this defines a sequence (x_n) contained in I. However, $x_n \to a$ as $n \to \infty$, but $f(x_n) \not\to f(a)$ as $n \to \infty$, which contradicts sequential continuity of f at $a \in I$. $\square$

Remark. To prove discontinuity of $f : I \to \mathbb{R}$ at $a \in I$ where $I \subset \mathbb{R}$ is an open interval, it therefore suffices to find *one* sequence (x_n) in I such that $x_n \to a$ but $f(x_n) \not\to f(a)$.

Example 1.18. The function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is not continuous at 0.



The idea is that for any $\delta > 0$, the function f oscillates infinitely many times on the interval $(0, \delta)$, so in particular we can take a sequence converging to zero where the function is constantly 1, which is not equal to $f(0) = 0$.

Let $x_n = \frac{1}{2\pi n + \frac{\pi}{2}}$. Then $x_n \to 0$, but $f(x_n) = \sin\left(2\pi n + \frac{\pi}{2}\right) = 1$ for all $n \in \mathbb{N}$, so $f(x_n) \to 1 \neq 0 = f(0)$, so f is discontinuous at 0.

Exercise. Show that the function $\chi_{\mathbb{Q}} : \mathbb{R} \to \mathbb{R}$ defined by

$$\chi_{\mathbb{Q}}(x) := \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

is discontinuous everywhere. *(Hint: use the sequential definition of density.)*

1.2.3 The Extreme Value Theorem

We now define continuity on closed and bounded intervals (i.e. intervals of the form $[b, c]$ for some $b < c$ in $\mathbb{R}$. Before, we were only working with open intervals. The definition for continuity on such an interval is a natural extension of continuity on open intervals:

Definition 1.19. Let $K = [b, c]$ be a closed and bounded interval for some $b < c$ in $\mathbb{R}$ and let $f : K \to \mathbb{R}$ be a function. We say that f is *continuous* on K (or just *continuous*) if f is continuous on (b, c) and also

$$\lim_{x\to b^+} f(x) = f(b), \qquad \lim_{x\to c^-} f(x) = f(c).$$

Lemma 1.20. *Let $I \subset \mathbb{R}$ be an open interval and let $K \subset I$ be a closed and bounded interval. If $f : I \to \mathbb{R}$ is continuous, then $f|_K : K \to \mathbb{R}$ is continuous.*

Proof. Follows from the definition. □

In practice, this is mostly how we will produce continuous functions on closed bounded intervals.

It is not hard to check that all our previous results about continuous functions still apply to functions defined on a closed and bounded interval K, the proofs need only trivial modifications.

Definition 1.21. Let $D \subset \mathbb{R}$ be any subset of $\mathbb{R}$ and let $f : D \to \mathbb{R}$ be a function. We say that f is *bounded* if the set $\{f(x) : x \in D\}$ is bounded, i.e. there exists $M > 0$ such that $|f(x)| \leq M$ for all $x \in D$.

Lemma 1.22. *Let $K = [b, c]$ be a closed and bounded interval. If $f : K \to \mathbb{R}$ is continuous, then f is bounded.*

Proof. Suppose for contradiction that f is not bounded. Then for each $M > 0$, there exists $x \in K$ such that $f(x) > M$. In particular, for $n \in \mathbb{N}$ define $M_n = n$. Then by definition we can construct a sequence (x_n) in K inductively such that $f(x_n) > n$ for all $n \in \mathbb{N}$.

Now, (x_n) is a bounded, since $x_n \in K$ for all $n \in \mathbb{N}$, so by the Bolzano-Weierstraß Theorem in $\mathbb{R}$ (Theorem **??**), there exists a convergent subsequence (x_{n_j}) with $\lim_{j\to\infty} x_{n_j} = L$. However, for all $j \in \mathbb{N}$, $b \leq x_{n_j} \leq c$, so by the Order Limit Theorem (Theorem **??**), $b \leq L \leq c$, so $L \in K$. By the sequential characterisation of continuity in $\mathbb{R}$ (Theorem 1.17), it follows that $\lim_{j\to\infty} f(x_{n_j}) = f(L)$. However, this is a contradiction because by construction, for each $j \in \mathbb{N}$, $f(x_{n_j}) > n_j \geq j$, so $f(x_{n_j}) \to \infty$ as $j \to \infty$, so it cannot converge. Therefore, f is bounded. □

Remark. This is essentially a topological result. Compact sets are closed (we will define this precisely later, but this means for any convergent sequence in the set, the limit cannot "escape" the set, i.e. cannot lie outside of the set), and the set is bounded. Intuitively, this means there is “no escape” for the function, if it is continuous (and therefore preserves this compact property).

If the interval is open, say $(0, 1)$, the function $f(x) = 1/x$ is continuous but unbounded because a convergent sequence $x_n \in (0, 1)$ can get arbitrarily close to 0 without ever equalling zero, allowing the function to blow up at 0. If the interval is unbounded, say $[0, \infty)$, $f(x) = x$ is continuous but unbounded. A sequence (x_n) contained in $[0, \infty)$ can escape to infinity, allowing the function to escape to infinity also. Compactness means that the *continuous* function's values have nowhere to esacpe to, because continuity preserves compactness, as we will see later. Below, we prove this for compact *intervals*, though continuity also preserves compact sets in general.

Definition 1.23. Let $D \subset \mathbb{R}$ be any subset of $\mathbb{R}$ and let $f : D \to \mathbb{R}$ be a bounded function. We define

$$\sup_{x\in D} f(x) := \sup\{f(x) : x \in D\} \quad \text{and} \quad \inf_{x\in D} f(x) := \{f(x) : x \in D\}.$$

Theorem 1.24 (Extreme Value Theorem)**.** *Let $K = [b, c]$ be a closed, bounded (compact) interval, and let $f : K \to \mathbb{R}$ be continuous. Then f attains its supremum and infimum. That is, there exists $d, e \in K$ such that*

$$f(d) = \sup_{x\in K} f(x) \quad \textit{and} \quad f(e) = \inf_{x\in K} f(x).$$

Proof. We will prove that f attains its supremum; the proof for attaining its infimum is similar and is left as an exercise.

Write $s = \sup_{x\in K} f(x)$, and suppose for contradiction that f does *not* attain its supremum, i.e. $f(x) \neq s$ for all $x \in K$. Then the following function is well-defined:

$$g : K \to \mathbb{R}, \quad x \mapsto \frac{1}{s - f(x)}.$$

Since f is continuous, and $s - f(x) \neq 0$ for all $x \in K$, g is continuous by the Algebra for Continuous Functions (Theorem 1.12), so by Lemma 1.22, g is bounded.

However, by definition of supremum, for each $n \in \mathbb{N}$ there exists $x_n \in K$ such that $s - \frac{1}{n} < f(x_n) < s$. This implies that, for $n \in \mathbb{N}$,

$$g(x_n) = \frac{1}{s - f(x_n)} > \frac{1}{s - \left(s - \frac{1}{n}\right)} = n,$$

so $g(x_n) \to \infty$ as $n \to \infty$, which contradicts g bounded. □

1.2.4 The Intermediate Value Theorem

Theorem 1.25 (Intermediate Value Theorem)**.** *Let $K = [b, c]$ be a closed and bounded interval and suppose $f : K \to \mathbb{R}$ is continuous. Then*

1. *If $f(b) \leq f(c)$ and A is any real number such that $f(b) \leq A \leq f(c)$, then there exists $a \in K$ such that $f(a) = A$.*

2. *If $f(c) \leq f(b)$ and A is any real number such that $f(c) \leq A \leq f(b)$, then there exists $a \in K$ such that $f(a) = A$.*

Proof. We prove the first case. Define the set $S := \{x \in K : f(x) \leq A\}$. Since $f(b) \leq A$, $b \in S$ so S is non-empty, and $S \subseteq K$ so S is bounded. Therefore, by the completeness of $\mathbb{R}$, $a := \sup S$ exists. We claim that $f(a) = A$; suppose for contradiction that $f(a) \neq A$.

Case 1: $f(a) < A$. First we note that $f(c) \geq A$, but by assumption $f(a) < A$, so $a \neq c$, i.e. $a < c$. Since f is continuous, if we pick $\varepsilon = \frac{A-f(a)}{2}$, there exists $\delta > 0$ such that for any $x \in (a-\delta, a+\delta) \cap K$, we have

$$|f(x) - f(a)| < \varepsilon = \frac{A - f(a)}{2},$$

i.e. for $x \in (a-\delta, a+\delta) \cap K$, we have

$$f(x) < f(a) + \varepsilon = f(a) + \frac{A - f(a)}{2} = \frac{f(a) + A}{2} < A,$$

so $x \in (a-\delta, a+\delta) \cap K \implies x \in S$. In particular, since $a < c$, we can find a $\delta' > 0$ small enough such that $I := [a, a+\delta') \subset K$ and $x \in I \implies x \in S$ by the above argument. Therefore, picking any $t \in (a, a+\delta')$, we have that $t \in S$ but $t > a$, contradicting the fact that a is an upper bound for S.

Case 2: $f(a) > A$. First we note that $f(b) \leq A$ but by assumption $f(a) > A$, so $a \neq b$, i.e. $a > b$. Since f is continuous, if we pick $\varepsilon = \frac{f(a)-A}{2}$, there exists $\delta > 0$ such that for any $x \in (a-\delta, a+\delta) \cap K$, we have

$$|f(x) - f(a)| < \varepsilon = \frac{f(a) - A}{2},$$

i.e. for $x \in (a-\delta, a+\delta) \cap K$, we have

$$f(x) > f(a) - \varepsilon = f(a) - \frac{f(a) - A}{2} = \frac{f(a) + A}{2} > A,$$

so $x \in (a-\delta, a+\delta) \cap K \implies x \notin S$. In particular, since $a > b$, we can find a $\delta' > 0$ small enough such that $I := (a-\delta', a] \subset K$ and $x \in I \implies x \notin S$ by the previous argument. Therefore, for every $x > a - \delta'$, $f(x) > A$, so $S \subseteq [b, a-\delta']$, so $a - \delta'$ is an upper bound for S, contradicting that a is the least upper bound for S.

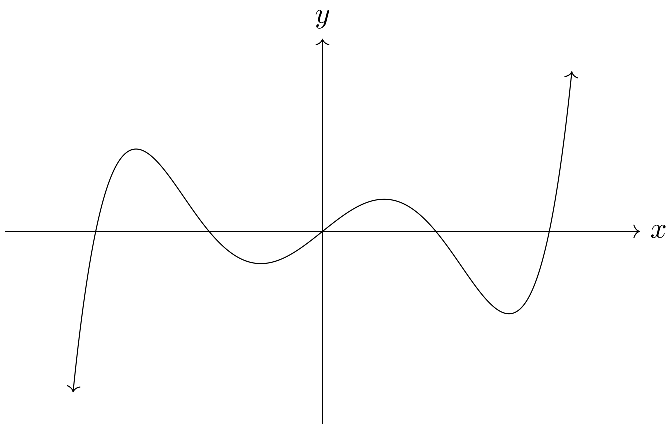
In both cases, we reach a contradiction, so $f(a) = A$.
To prove the second case, note that if $f(c) \leq f(b)$, and A is any real number such that $f(c) \leq A \leq f(b)$, then $-f(b) \leq -A \leq -f(c)$ and $-f$ is continuous by the Algebra for Continuous Functions (Theorem 1.12), so by the first case there exists $a \in K$ such that $-f(a) = -A$, i.e. $f(a) = A$. □

Example 1.26 (Fixed Point Application)**.** Let $f : [0,1] \to [0,1]$ be continuous. We can prove f has a fixed point (a point c where $f(c) = c$).

Define $g(x) = f(x) - x$. By the Algebra for Continuous Functions (Theorem 1.12), g is continuous. Since $f(0) \in [0,1]$, $g(0) = f(0) - 0 \geq 0$. Since $f(1) \in [0,1]$, $g(1) = f(1) - 1 \leq 0$. Since g is continuous, the Intermediate Value Theorem (IVT) guarantees a point $c \in [0,1]$ such that $g(c) = 0$, i.e. $f(c) = c$.

Example 1.27. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^5 + x - 1$. Then $f(0) = -1 < 0$ and $f(1) = 1 > 0$, so by the Intermediate Value Theorem there exists $c \in (0, 1)$ such that $f(c) = 0$, i.e. there is a root of f in $(0, 1)$.

In fact, we can generalise this result for odd degree polynomials. Intuitively, the tails of an odd degree polynomial must diverge in opposite directions to each other because of the odd leading power of x:



Corollary 1.28. *Let $P : \mathbb{R} \to \mathbb{R}$ be any polynomial of odd degree. Then P has at least one root.*

Proof. Let $P : \mathbb{R} \to \mathbb{R}$ be defined by

$$P(x) = a_{2n+1}x^{2n+1} + a_{2n}x^{2n} + \cdots + a_1 x + a_0,$$

where $a_i \in \mathbb{R}$ are constants with $a_{2n+1} \neq 0$, and n is some natural number. Without loss of generality, assume $a_{2n+1} > 0$ (if not, then replace P with $-P$ which has a root if and only if P does). We write $Q(x) = a_{2n}x^{2n} + \cdots + a_1 x + a_0$.

Consider evaluating Q on the natural numbers. Then $Q(i)/i^{2n+1} \to 0$ as $i \to \infty$, so if i is large enough then $|Q(i)/i^{2n+1}| < a_{2n+1}$ and so $|Q(i)| < a_{2n+1}i^{2n+1}$. Hence

$$P(i) = a_{2n+1}i^{2n+1} + Q(i) \geq a_{2n+1}i^{2n+1} - |Q(i)| > 0,$$

and

$$P(-i) \leq -a_{2n+1}i^{2n+1} + |Q(i)| < 0,$$

for sufficiently large i. Now fix j sufficiently large such that the above inequalities hold; then P is continuous on $[-j, j]$ with $P(-j) < 0$ and $P(j) > 0$, so P has a root in $(-j, j)$. $\square$

Theorem 1.29. *Let I be an interval. If $f : I \to \mathbb{R}$ is continuous and injective (one-to-one) on I, then f is either strictly increasing or strictly decreasing on I.*

1.3 Topology of $\mathbb{R}$

Definition 1.30 (ε-neighborhood)**.** Let $z \in \mathbb{C}$ and $\varepsilon > 0$. An *ε-neighborhood* of z is the set

$$V_\varepsilon(z) := \{w \in \mathbb{C} : |w - z| < \varepsilon\}.$$

Definition 1.31 (Open Set in $\mathbb{C}$)**.** A set $O \subset \mathbb{C}$ is *open* if for all $z \in O$, there exists $\varepsilon > 0$ such that $V_\varepsilon(z) \subset O$.

Remark. Intuitively, this means that if O is an open set, then every point has a small "neighborhood" that is entirely contained within O, i.e. the point has "room", it is not right at the boundary, say.

Example 1.32. For $a < b$ in $\mathbb{R}$, the open interval $(a, b) = \{x \in \mathbb{R} : a < x < b\}$ is an open set (exercise: show this), however $[a, b]$ is not a closed set.

Lemma 1.33. *Let $n \in \mathbb{N}$ and suppose $O_1, \ldots, O_n \subset \mathbb{C}$ are open. Then*

$$\bigcap_{i=1}^{n} O_i \subset \mathbb{C}$$

is open.

Proof. Put $O := \bigcap_{i=1}^{n} O_i$ and let $z \in O$ be arbitrary. Then for each i, there exists ε_i such that $V_{\varepsilon_i}(z) \subset O_i$. Take $\varepsilon := \min_i\{\varepsilon_i\}$. Then we have $V_\varepsilon(z) \subset O_i$ for each i, so $V_\varepsilon(z) \subset O$, so O is open. □

Exercise. Find an example of a collection of open sets $\{O_n\}_{n\in\mathbb{N}}$ in $\mathbb{R}$ such that $\bigcap_{n\in\mathbb{N}} O_n$ is not open.

Lemma 1.34. *Let $\{O_\alpha\}_{\alpha\in I}$ be an arbitrary collection of open sets in $\mathbb{C}$. Then*

$$\bigcup_{\alpha\in I} O_\alpha \subset \mathbb{C}$$

is open.

Proof. Exercise. □

Definition 1.35 (Closed Set in $\mathbb{C}$)**.** A subset $A \subset \mathbb{C}$ is *closed* if for every convergent sequence (a_n) contained in A we have

$$\lim_{n\to\infty} a_n \in A.$$

Lemma 1.36. *Let $A_1, \ldots, A_n \subset \mathbb{C}$ be closed. Then*

$$\bigcup_{i=1}^{n} A_i \subset \mathbb{C}$$

is closed.

Proof. Exercise. □

Lemma 1.37. *Let $\{A_\alpha\}_{\alpha\in I}$ be an arbitrary collection of closed sets in $\mathbb{C}$. Then*

$$\bigcap_{\alpha\in I} A_\alpha \subset \mathbb{C}$$

is closed.

Proof. Exercise. □

Exercise. Find an example of a collection of closed sets $\{A_n\}_{n\in\mathbb{N}}$ in $\mathbb{R}$ such that $\bigcup_{n\in\mathbb{N}} A_n$ is not closed.

Definition 1.38 (Bounded Set in $\mathbb{C}$)**.** A subset $B \subset \mathbb{C}$ is *bounded* if there exists $R \in \mathbb{R}_{>0}$ such that

$$B \subset \{z \in \mathbb{C} : |z| \leq R\}.$$

Definition 1.39 (Compact Set in $\mathbb{C}$)**.** A subset $K \subset \mathbb{C}$ is *compact* if

1. K is closed;
2. K is bounded.

Lemma 1.40. *Let $K_1, \ldots, K_n \subset \mathbb{C}$ be compact. Then*

$$\bigcup_{i=1}^{n} K_i \subset \mathbb{C}$$

is compact. Moreover, the intersection of an arbitrary collection of compact subsets of $\mathbb{C}$ is compact.

Proof. Exercise. □

There is a useful characterisation of compact sets using the Bolzano-Weierstraß Theorem:

Theorem 1.41. *A subset $K \subset \mathbb{C}$ is compact if and only if for every sequence (z_n) in K, there exists a convergent subsequence (z_{n_j}) such that*

$$\lim_{j\to\infty} z_{n_j} \in K.$$

Proof. Suppose $K \subset \mathbb{C}$ is compact, and let (z_n) be an arbitrary sequence in K. Since K is bounded, (z_n) is bounded so by the Bolzano-Weierstraß Theorem in $\mathbb{C}$ (Theorem **??**), there exists a convergent subsequence (z_{n_j}) of (z_n) such that $\lim_{j\to\infty} z_{n_j} =: z$ exists. Since K is closed, $z \in K$.

Conversely, suppose that for every sequence (z_n) in K, there exists a convergent subsequence (z_{n_j}) such that $z := \lim_{j\to\infty} z_{n_j} \in K$. We now need to argue that K is closed and bounded.

K bounded: suppose for contradiction that K is unbounded. Then we could choose a sequence (z_n) in K such that $|z_n| \to \infty$ as $n \to \infty$, which has no convergent subsequence in $\mathbb{C}$, contradicting our assumption, so K must be bounded.

K closed: let (z_n) be a convergent sequence in K such that $z_n \to z$ as $n \to \infty$. We need to show that $z \in K$. By our assumption, there exists a convergent subsequence (z_{n_j}) of (z_n) converging to some $w \in K$. However, convergent sequences have unique limits so $z = w$ and thus $z \in K$, so K is closed.

Therefore, K is compact. □

Theorem 1.42 (Continuity preserves compact sets)**.** *Let $K \subset \mathbb{C}$ and $f : K \to \mathbb{C}$ be continuous. Then the image*

$$f(K) = \{f(z) : z \in K\} \subset \mathbb{C}$$

is compact.

Proof. Let (w_n) be a sequence in $f(K)$ with, and let (z_n) be the sequence such that for all $n \in \mathbb{N}$, $f(z_n) = w_n$. Since K is compact, by Theorem 1.41, there exists a convergent subsequence (z_{n_j}) of (z_n) such that $\lim_{j\to\infty} z_{n_j} = z$ for some $z \in K$. By the sequential characterisation of continuity, it follows that

$$\lim_{j\to\infty} w_{n_j} = \lim_{j\to\infty} f(z_{n_j}) = f(z),$$

so (w_n) has a convergent subsequence (w_{n_j}) whose limit $f(z)$ is in K, so by Theorem 1.41 again, $f(K)$ is compact. □

Theorem 1.43. *Let $K \subset \mathbb{C}$ be compact and let $f : K \to \mathbb{R}$ be continuous. Then there exist $z_1, z_2 \in K$ such that*

$$f(z_1) \le f(z) \le f(z_2) \quad \forall z \in K.$$

In other words, f attains both its minimum and maximum on K.

Proof. By Theorem 1.42, $f(K)$ is compact, so in particular $f(K)$ is bounded, so the real numbers

$$m := \inf f(K), \quad M := \sup f(K)$$

exist. Since $f(K)$ is closed as well, we have $m, M \in K$ (to see this, note that for any bounded set $S \subset \mathbb{R}$, by definition of supremum and infimum, there are convergent sequences (x_n) and (y_n) such that $x_n \to \inf S$ and $y_n \to \sup S$, so for compact sets $\inf S$, $\sup S$ must be in S). Therefore, there exist $z_1, z_2 \in K$ such that $f(z_1) = m$ and $f(z_2) = M$, which completes the proof. □

The statement of Theorem 1.43 cannot be directly generalised to functions with codomain $\mathbb{C}$. This is because the complex numbers do not possess a natural total ordering compatible with field operations. Inequalities such as $f(z_1) \le f(z) \le f(z_2)$ are undefined when the outputs are complex numbers; therefore, a complex-valued function cannot attain a "maximum" or "minimum" value in the traditional sense (think about a circle in the complex plane; every point on this circle has equal magnitude).

However, the topological core of the theorem—that continuous functions map compact sets to compact sets—does apply. We can use this to prove a highly useful variation of the Extreme Value Theorem in $\mathbb{C}$ by looking at the *modulus* (magnitude) of the complex function:

Theorem 1.44 (Extreme Value Theorem in $\mathbb{C}$)**.** *Let $K \subset \mathbb{C}$ be compact and $f : K \to \mathbb{C}$ be continuous. Then $f(K)$ is bounded in $\mathbb{C}$, and there exist $z_1, z_2 \in K$ such that*

$$|f(z_1)| \leq |f(z)| \leq |f(z_2)| \quad \forall z \in K.$$

Proof. First we note that by Theorem 1.42, $f(K)$ is compact, so in particular is bounded. Now, define $h : K \to \mathbb{R}$ by taking the modulus of our original function:

$$h(z) = |f(z)|$$

The function h can be viewed as the composition of f and the absolute value function $g : \mathbb{C} \to \mathbb{R}$ defined by $g(w) = |w|$. We are given that $f : K \to C$ is continuous. Furthermore, the absolute value function $g : \mathbb{C} \to \mathbb{R}$ is continuous (this follows from the reverse triangle inequality, $\big||w_1| - |w_2|\big| \leq |w_1 - w_2|$). Since the composition of continuous functions is continuous (by Theorem 1.14), $h(z)$ is continuous on K.

By Theorem 1.43, there exist $z_1, z_2 \in K$ such that:

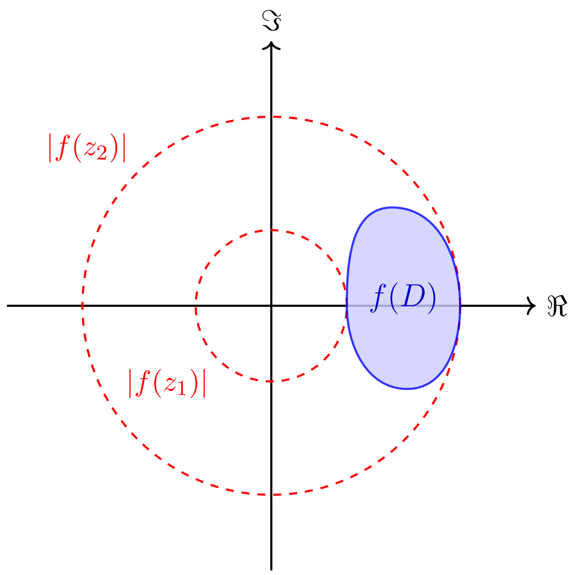
$$h(z_1) \leq h(z) \leq h(z_2) \quad \text{for all } z \in K$$

Substituting $|f(z)|$ back in for $h(z)$, we obtain the desired inequality for the extreme values of the modulus:

$$|f(z_1)| \leq |f(z)| \leq |f(z_2)| \quad \text{for all } z \in K.$$

□

Remark. Both the compactness of K and the continuity of f are essential hypotheses of this theorem.



Example 1.45. The following examples satisfy both hypotheses of the Extreme Value Theorem in $\mathbb{C}$ (Theorem 1.44): the domain $K \subset \mathbb{C}$ is compact (closed and bounded), and the function f is continuous.

1. Let $f(z) = z^2$ defined on the closed unit disk, $K = \{z \in \mathbb{C} : |z| \leq 1\}$. Because K is closed and bounded, it is compact. The polynomial f is continuous everywhere. The modulus is $|f(z)| = |z^2| = |z|^2$. The minimum modulus is 0, attained at $z_1 = 0$. The maximum modulus is 1, attained anywhere on the boundary (e.g., $z_2 = 1$ or $z_2 = i$). The image $f(K)$ is the closed unit disk itself, which is bounded.

2. Let $f(z) = e^z$ defined on the rectangle $K = \{x+iy : 0 \leq x \leq 1, 0 \leq y \leq \pi\}$. K is compact and f is continuous. The modulus is $|f(z)| = |e^{x+iy}| = |e^x e^{iy}| = e^x$. The minimum modulus is 1 (attained when $x = 0$, anywhere on the left edge), and the maximum modulus is e (attained when $x = 1$, anywhere on the right edge). The image $f(K)$ is bounded between radius 1 and e.

Example 1.46. The following examples demonstrate why both hypotheses in Theorem 1.44 are strictly required.

1. Let $D = (0, 1)$ and $f(x) = x$. The domain is not compact because it does not contain its limit points. The function is continuous, but it never attains a maximum or minimum on this open interval.

2. Let $D = [1, \infty)$ and $f(x) = x$. The domain is closed but extends infinitely. The function is continuous, but the set $f(D)$ is unbounded, meaning no maximum exists.

3. Let $D = [0, 1]$. Define $f(x) = 1/x$ for $x \in (0, 1]$, and $f(0) = 0$. The domain is compact. However, due to the discontinuity at $x = 0$, the function approaches infinity as $x \to 0$. It does not attain a maximum value.

4. Let $D = \{z \in \mathbb{C} : |z| \leq 1\}$ (the closed unit disk). Define $f(z) = 1/z$ for $z \neq 0$, and set $f(0) = 0$. The domain is compact, but the function is discontinuous at the origin. The magnitude $|f(z)| = 1/|z|$ grows arbitrarily large near 0, so $f(D)$ is unbounded and no maximum modulus exists.

1.4 Uniform Continuity

We now consider a *stronger* notion of continuity, which controls the function's behaviour *globally*, rather than *locally*, like the notion of "normal" continuity we have just been studying (left and right limits matching the function value at a point).

Definition 1.47 (Uniform Continuity)**.** Let $D \subset \mathbb{C}$. A function $f : D \to \mathbb{C}$ is *uniformly continuous* on D if

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x, y \in D,\ |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

Remark. Normal continuity is a local property (defined *at a point* a). The δ depends on both the tolerance ε *and* the specific location a (i.e., we may write $\delta = \delta(\varepsilon, a)$).

If a function gets infinitely steep (like $1/x$ near 0), you need smaller and smaller δ's as you move, to stay within the same ε tolerance.

Uniform continuity guarantees that the function does not suddenly oscillate very fast when you move to a different part of the domain.

Uniform continuity is a global property (defined *on a set*). It means you can find a single "universal δ" that works for the given ε everywhere on the domain (i.e. we may write $\delta = \delta(\varepsilon)$).

Example 1.48. The function $f : (0,1) \to \mathbb{R}$ defined by $f(x) = \frac{1}{x}$ is not uniformly continuous.

Intuitively, near 0 the graph of $f(x) = \frac{1}{x}$ becomes extremely steep: very small changes in x can cause very large changes in $\frac{1}{x}$, suggesting that no single $\delta > 0$ can work for all points in $(0,1)$ at once.

Indeed, pick $\varepsilon := \frac{1}{2}$, and let $\delta > 0$ be arbitrary. Let $n \in \mathbb{N}$ be large enough such that $0 < \frac{1}{n} < \delta$, and pick $x = \frac{1}{n}$, $y = \frac{1}{2n}$. Then we have $|x - y| = \frac{1}{n} < \delta$, but

$$|f(x) - f(y)| = |n - 2n| = n \geq 1 > \frac{1}{2} = \varepsilon,$$

so f is not uniformly continuous on $(0,1)$.

Again, the problem in the above example was that f diverges to ∞ at 0 and the domain $(0,1)$ is not closed, i.e. we were allowed to take points arbitrarily close to 0, allowing the function to grow too rapidly to control.

Therefore, given our brief discussion of compact sets in the previous section, it is natural to suspect that continuity on a compact set is enough to be uniformly continuous. Indeed, we have the following theorem:

Theorem 1.49 (Heine's Theorem). *Let $K \subset \mathbb{C}$ be compact and let $f : K \to \mathbb{C}$ be continuous. Then f is uniformly continuous on K.*

Proof. Suppose for contradiction that f is not uniformly continuous on K. Let $\varepsilon_0 > 0$ be such that for any $\delta > 0$, there exist $x, y \in K$ such that $|x - y| < \delta$ and $|f(x) - f(y)| \geq \varepsilon_0$.

In particular, for $n \in \mathbb{N}$ we let $\delta_n := \frac{1}{n}$, and for $n \in \mathbb{N}$ we inductively choose $z_n, w_n \in K$ such that $|z_n - w_n| < \delta_n$ and $|f(z_n) - f(w_n)| \geq \varepsilon_0$ by definition.

Since K is compact, (z_n) has a convergent subsequence (z_{n_j}) such that $z := \lim_{j\to\infty} z_{n_j} \in K$. Furthermore, we have

$$|z_{n_j} - w_{n_j}| < \frac{1}{n_j} \leq \frac{1}{j} \to 0$$

as $j \to \infty$, so $w_{n_j} \to z$ as $j \to \infty$.

Now, by the sequential characterisation of continuity (Theorem 1.17), since f is continuous we get

$$\lim_{j\to\infty} f(z_{n_j}) = f(z) \quad \text{and} \quad \lim_{j\to\infty} f(w_{n_j}) = f(z),$$

so in particular

$$\lim_{j\to\infty} |f(z_{n_j}) - f(w_{n_j})| = 0.$$

Therefore, for sufficiently large j, we have

$$|f(z_{n_j}) - f(w_{n_j})| < \frac{\varepsilon_0}{2} < \varepsilon_0,$$

which contradicts the construction of (z_n) and (w_n), which satsify $|f(z_n) - f(w_n)| \geq \varepsilon_0$ for all $n \in \mathbb{N}$. $\square$

Definition 1.50 (Lipschitz Continuity)**.** Let $D \subset \mathbb{R}$. A function $f : D \to \mathbb{R}$ is *Lipschitz continuous* if there exists a real constant $L \geq 0$ (the *Lipschitz constant*) such that for all $x, y \in D$,

$$|f(x) - f(y)| \leq L|x - y|.$$

Lipschitz continuity places a strict limit on how "steep" the function can get on D. Indeed, if $x < y$, the Lipschitz condition above rearranges to

$$\left|\frac{f(x) - f(y)}{x - y}\right| \leq L,$$

which places a uniform bound L on the average slope of the function on any interval (x, y)). If a function is differentiable, bounding its derivative ($|f'(x)| \leq L$) guarantees it is Lipschitz (exercise: once you have read the derivative chapter, prove this).

Remark. Lipschitz $\implies$ Uniformly Continuous $\implies$ Continuous, however *none* of the reverse implications hold (exercise: find counterexamples).

Example 1.51. The function $f : [0, 1] \to \mathbb{R}$ defined by $f(x) = \sqrt{x}$ is continuous. Because $[0, 1]$ is compact, it is uniformly continuous by Heine's Theorem. However, it is *not* Lipschitz continuous because its derivative $\frac{1}{2\sqrt{x}}$ approaches infinity near $x = 0$, meaning its slope cannot be globally bounded by a single constant L. To prove this rigorously (since we have not defined derivative yet), we argue as follows.

The idea is to fix one point at 0 and let (x_n) be a sequence converging to 0, to get the slope diverging as $n \to \infty$ (visually, imagine the tangent line at a point $x_0 > 0$ getting steeper and steeper as $x_0 \to 0$).

Suppose for contradiction there exists $L \geq 0$ such that for all $x, y \in [0, 1]$,

$$|\sqrt{x} - \sqrt{y}| \leq L|x - y|.$$

Let $y = 0$, and for $n \in \mathbb{N}$ define $x_n := \frac{1}{n^2}$. Given $n \in \mathbb{N}$, we have by the Lipschitz condition,

$$\left|\sqrt{\frac{1}{n^2}} - 0\right| = \frac{1}{n} \leq L\left|\frac{1}{n^2} - 0\right|,$$

so rearranging we get $n \leq L$ for all $n \in \mathbb{N}$, which is a contradiction (e.g. pick $n = \lceil L \rceil + 1$ since n was arbitrary). Therefore, f is not Lipschitz on $[0, 1]$.